Algebra 1 Name___________________
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Lesson 1 Practice Period___________________

1. The equations $y = -3(x+2)(x-3)$ and $y = -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$ both represent the same quadratic. Rewrite both equations in standard form, $y = ax^2 + bx + c$. Show your work.

$y = -3(x+2)(x-3)$	$y = -3\left(x-\frac{1}{2}\right)^2 + \frac{75}{4}$
$y = -3(x^2 - 3x + 2x - 6)$ $y = -3(x^2 - x - 6)$ $y = -3x^2 + 3x + 18$	$y = -3\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2}\right) + \frac{75}{4}$ $y = -3\left(x^2 - x + \frac{1}{4}\right) + \frac{75}{4}$ $y = -3x^2 + 3x - \frac{3}{4} + \frac{75}{4}$ $y = -3x^2 + 3x + 18$

2. The equations $y = -(x+6)(x-3)$ and $y = -(x-(-6))(x-3)$ are equivalent. How can both x-intercepts be determined if the equation $y = -(x+6)(x-3)$ is given?

The x-intercepts can be determined by using Zero Product Property. Set each factor equal to zero and then solve to find the x-intercepts.

3. The equation $y = -2x^2 - 4x + 30$ can be used to determine the y-intercept. Show how each of the three forms can be used to find the y-intercept.

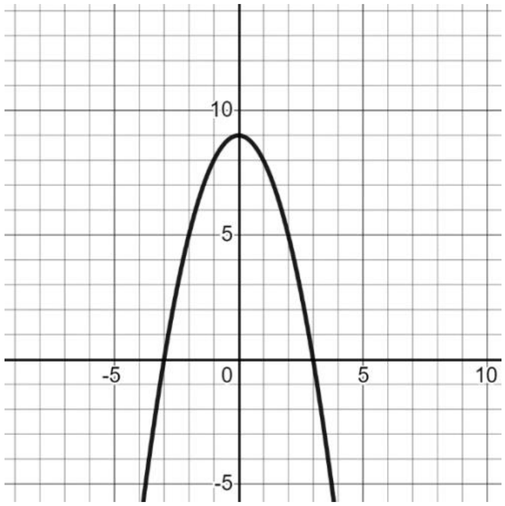
$y = -2x^2 - 4x + 30$	$y = -2(x+5)(x-3)$	$y = -2(x+1)^2 + 32$
$y = -2(0)^2 - 4(0) + 30$ $y = -2(0) - 4(0) + 30$ $y = 30$	$y = -2(0+5)(0-3)$ $y = -2(5)(-3)$ $y = 30$	$y = -2(0+1)^2 + 32$ $y = -2(1)^2 + 32$ $y = -2 + 32$ $y = 30$

4. A table of values for $y = \frac{1}{4}(x-2)^2 + 8$ is shown. What features can be found using the table of values?

- ☐ x-intercepts
- ☑ y –intercepts
- ☑ vertex
- ☑ axis of symmetry
- ☑ increasing/decreasing
- ☐ positive/negative

x	y
-3	14.25
-2	12
-1	10.25
0	9
1	8.25
2	8
3	8.25

5. The graph of a quadratic equation is shown. Use words, ordered pairs, or symbols to describe the location of each key feature.



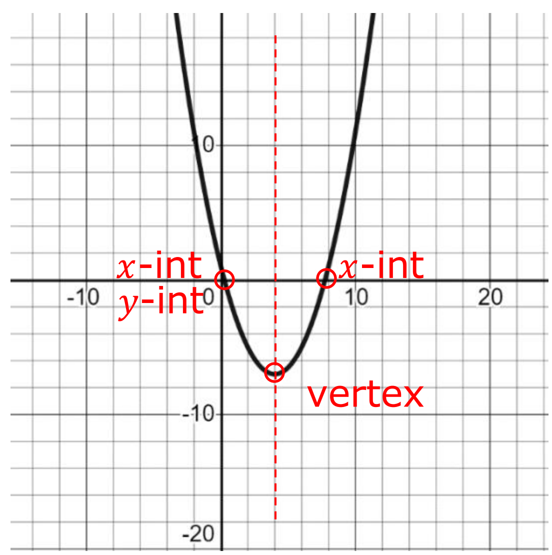
Key Feature	Location of Key Feature
x-intercepts	$(-3,0)$ $(3,0)$
y-intercepts	$(0,9)$
vertex	$(0,9)$
axis of symmetry	$x = 0$
increasing	$x < 0$
decreasing	$x > 0$

6. A table of values for $y = (x-1)^2$ is shown. What features can be found using the table of values?

- ☑ x –intercepts
- ☑ y –intercepts
- ☑ vertex
- ☑ axis of symmetry
- ☑ increasing/decreasing
- ☑ positive/negative

x	y
-1	4
0	1
1	0
2	1
3	4

Use the graph of the quadratic equation for questions 7-10.



7. Describe where the parabola is increasing.
$x > 4$

8. Describe where the parabola is decreasing.
$x < 4$

9. Draw the axis of symmetry for the parabola on the graph.

10. Circle the vertex, x –intercepts, and y – intercepts of the parabola. Label each point as vertex, x –intercepts, and y – intercepts.

Algebra 1
Unit 8
Lesson 2 Practice

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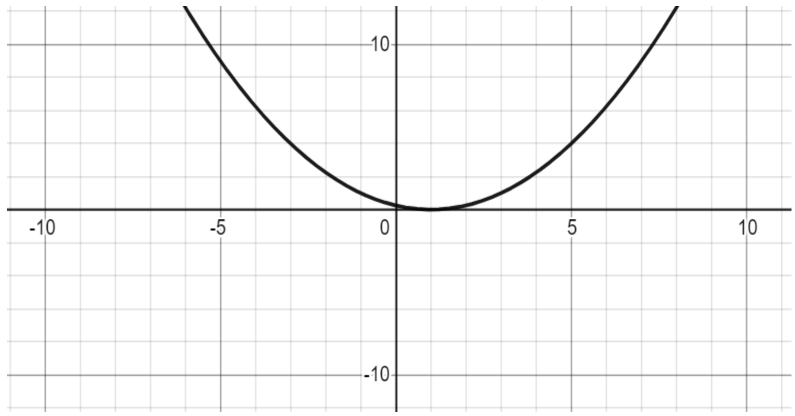
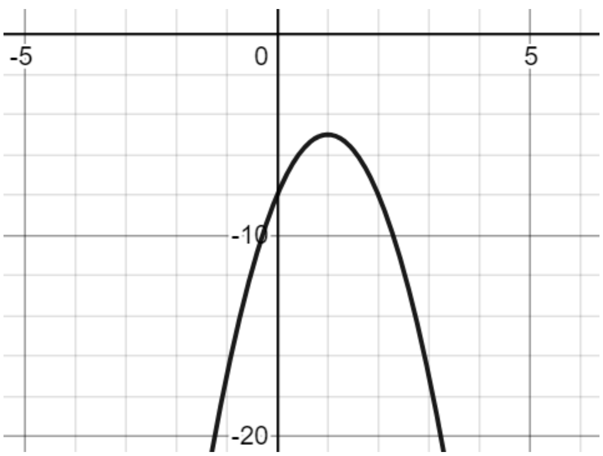


1. As the x-values of the parabola get infinitely *larger*, the y-values of the parabola approach
   - [x] ∞
   - [ ] $-\infty$

2. As the x-values of the parabola get infinitely *smaller*, the y-values of the parabola approach
   - [x] ∞
   - [ ] $-\infty$

Use the graph of the quadratic for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic.
   $-\infty < x < \infty$

4. Use words or symbols to describe the range of the quadratic.
   $-\infty < y \leq -5$

5. Determine the x-intercepts and y-intercepts of a quadratic using the equation $y = -\frac{4}{3}(x-2)(x+3)$.

$0 = \frac{-4}{3}(x-2)(x+3)$
$0 = (x-2)(x+3)$

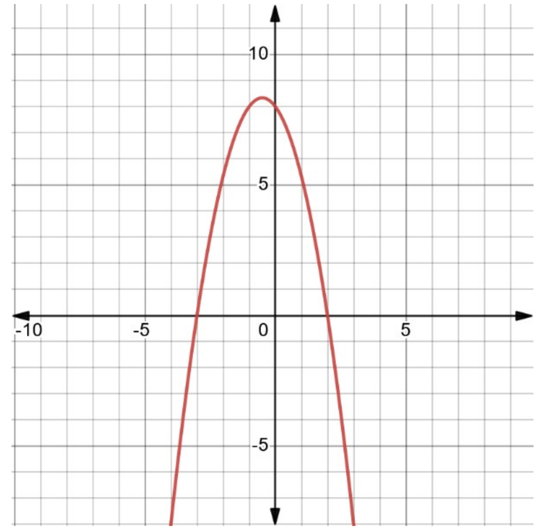
$x = 2$
$x = -3$
$(2,0)(-3,0)$

$y = \frac{-4}{3}(0-2)(0+3)$
$y = \frac{-4}{3}(-2)(3)$
$y = 8 \quad (0,8)$

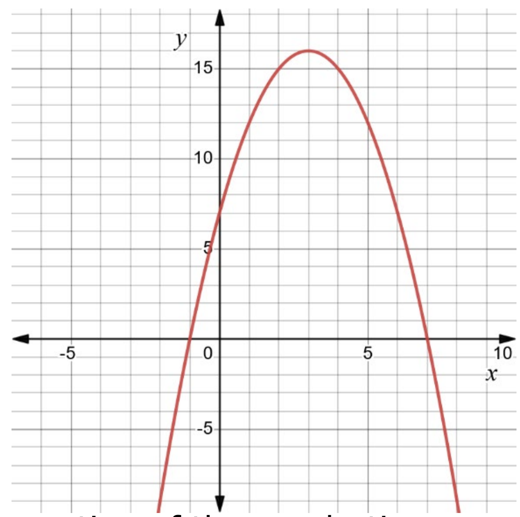
6. Use the table of values to determine the vertex of the parabola.
   Vertex $= (-0.5, 8.333)$

x	y
-2	5.333
-1.5	7
-1	8
-0.5	8.333
0	8
0.5	7
1	5.333

7. Use the key features you determined in question 5 and 6 to graph the parabola of the quadratic function.



8. Use the equation $y = -(x-7)(x+1)$ to graph the quadratic.



$0 = -(x-7)(x+1)$
$0 = (x-7)(x+1)$
$x = 7 \quad x = -1$
$(7,0) \quad (-1,0)$

$y = -(0-7)(0+1)$
$0 = -(-7)(1)$
$y = 7$
$(0,7)$

x	y
1	12
2	15
3	16
4	15

Vertex $= (3,16)$

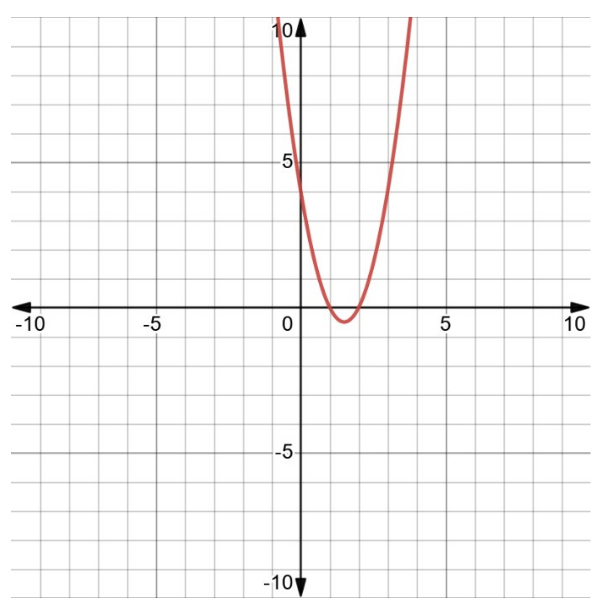
9. Determine all of the key features possible using the equation of the quadratic $y = 2(x-2)(x-1)$.

$0 = 2(x-2)(x-1)$
$0 = (x-2)(x-1)$
$x = 2 \quad x = 1$
$(2,0) \quad (1,0)$
x-intercepts

$y = 2(0-2)(0-1)$
$y = 2(-2)(-1)$
$y = 4$
$(0,4)$
y-intercept

x	y	
0	4	
0.5	1.5	
1	0	
1.5	-0.5	vertex, axis of symmetery
2	0	

10. Use the key features found in question 9 to graph the quadratic.



Algebra 1
Unit 8
Lesson 3 Practice

Name Answer Key______________
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Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y=(x-9)(x+12)$	○ Vertex form ✗ Factored form ○ Standard form	y-intercept x-intercepts Direction graph opens	$(0,-108)$ $(9,0)$ $(-12,0)$ Graph opens up $A>0$
2. $y=(x-5)^2-8$	✗ Vertex form ○ Factored form ○ Standard form	y-intercept Vertex Axis of symm Direction of graph	$(0,17)$ $(5,-8)$ $x=5$ Opens up $A>0$
3. $y=-\frac{1}{4}x^2-4x-4$	○ Vertex form ○ Factored form ✗ Standard form	y-intercept Direction of graph	$(0,-4)$ Opens down $A<0$
4. $y=7(x-1)^2$	✗ Vertex form ○ Factored form ○ Standard form	y-intercept Vertex Axis of symm Direction of graph	$(0,7)$ $(1,0)$ $x=1$ Opens up $A>0$
5. $y=-x^2+12x+36$	○ Vertex form ○ Factored form ✗ Standard form	y-intercept Direction of graph	$(0,36)$ Opens down $A<0$

6. Rewrite the quadratic equation in factored form.

$$y=x^2-3x-10$$
$$y=(x-5)(x+2)$$

7. Rewrite the quadratic equation in standard form.

$$y = (x - 3)^2 - 7$$

$$y = (x - 3)(x - 3) - 7$$

$$y = x^2 - 6x + 9 - 7$$

$$y = x^2 - 6x + 2$$

8. Rewrite the equation in factored form.

$$y = -x^2 + 11x - 28$$

$$y = -(x^2 - 11x + 28)$$

$$y = -(x - 7)(x - 4)$$

9. Rewrite the equation in standard form.

$$y = -2(x - 1)(x + 9)$$

$$y = -2(x^2 + 8x - 9)$$

$$y = -2x^2 - 16x + 18$$

10. Cyrus makes model rockets with his boy scout troop. The rockets launch straight up from 25 meters above the ground with a velocity of 20 meters/second. The height of the rocket is represented by the equation $y = -5x^2 + 20x + 25$, where x is time in seconds. Rewrite the equation in factored form.

$$y = -5(x^2 - 4x - 5)$$

$$y = -5(x - 5)(x + 1)$$

Algebra 1
Unit 8
Lesson 4 Practice

Name Answer Key ______________
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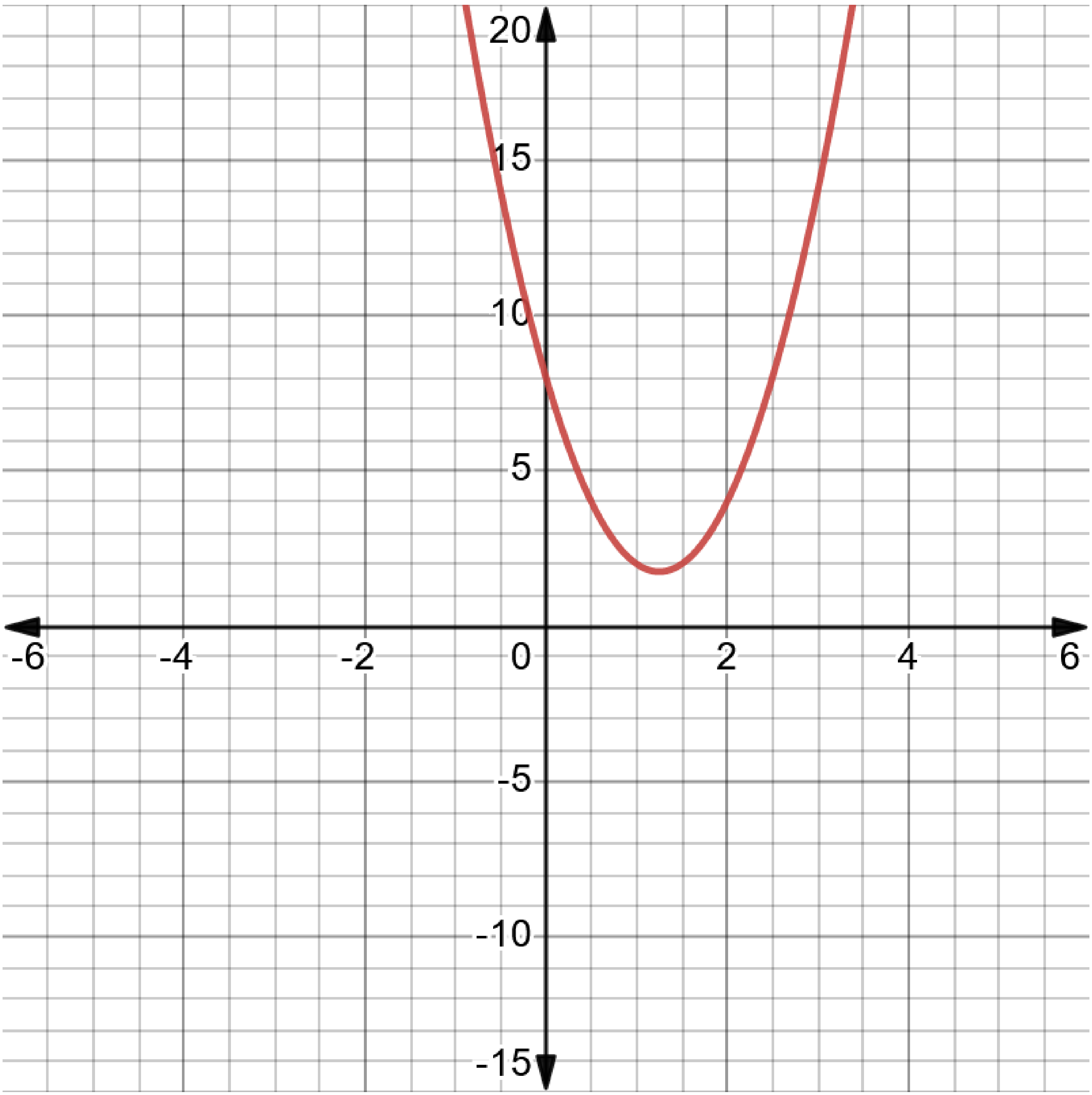
1. Graph the equation $y = 4x^2 - 10x + 8$.



2. Write the key features that can be determined from the equation.

$y = 4x^2 - 10x + 8$	
x-intercept	
y-intercept	$(0,8)$
Vertex	
Domain	All Real Numbers
Range	
Left end behavior	$x \to -\infty\,,\quad y \to \infty$
Increasing	
Decreasing	
Positive	
Negative	
Opens up/down	Up
Axis of symmetry	
Right end behavior	$x \to \infty,\quad y \to \infty$

Algebra 1 Name Answer Key ____________
Unit 8 Date____________________
Lesson 5 Practice Period____________________

Re-write each expression into vertex form by completing the square.

1. $x^2 - 10x + 1$
$[x^2 - 10x + 25] + 1 - 25$
$(x - 5)(x - 5) - 24$
$(x - 5)^2 - 24$

2. $m^2 + 8m - 11$
$[m^2 + 8m + 16] - 11 - 16$
$(m + 4)(m + 4) - 27$
$(m + 4)^2 - 27$

3. $x^2 - 6x + 9$
$x^2 - 6x + 9$
$(x - 3)(x - 3)$
$(x - 3)^2$

4. $j^2 - 2j - 3$
$[j^2 - 2j + 1] - 3 - 1$
$(j - 1)(j - 1) - 4$
$(j - 1)^2 - 4$

5. $x^2 + 8x - 10$
$[x^2 + 8x + 16] - 10 - 16$
$(x + 4)(x + 4) - 26$
$(x + 4)^2 - 26$

6. $h^2 + 2h - 3$
$[h^2 + 2h + 1] - 3 - 1$
$(h + 1)(h + 1) - 4$
$(h + 1)^2 - 4$

7. $q^2 - 12q + 26$

$[q^2 - 12q + 36] + 26 - 36$

$(q - 6)(q - 6) - 10$

$(q - 6)^2 - 10$

8. $d^2 - 2d - 8$

$[d^2 - 2d + 1] - 8 - 1$

$(d - 1)(d - 1) - 9$

$(d - 1)^2 - 9$

9. $y^2 + 12x + 20$

$[y^2 + 12x + 36] + 20 - 36$

$(y + 6)(y + 6) - 16$

$(y + 6)^2 - 16$

10. $u^2 + 16u - 22$

$[u^2 + 16u + 64] - 22 - 64$

$(u + 8)(u + 8) - 86$

$(u + 8)^2 - 86$

Algebra 1
Unit 8
Lesson 6 Practice

Name Answer Key ___________
Date_____________________
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Re-write each expression into vertex form by completing the square.

1. $4t^2 - 16t + 12$
$4(t^2 - 4t + 4) + 12 - 16$
$4(t - 2)^2 - 4$

2. $-2x^2 + 8x - 18$
$-2(x^2 - 4x + 4) - 18 + 8$
$-2(x - 2)^2 - 10$

3. $3r^2 - 18r - 12$
$3(r^2 - 6r + 9) - 12 - 27$
$3(r - 3)^2 - 39$

4. $-5h^2 - 20h + 3$
$-5(h^2 + 4h + 4) + 3 + 20$
$-5(h + 2)^2 + 23$

5. $3y^2 - 24y - 4$
$3(y^2 - 8x + 16) - 4 - 48$
$3(y - 4)^2 - 52$

6. $-2u^2 + 12u + 7$
$-2(u^2 - 6u + 9) + 7 + 18$
$-2(u - 3)^2 + 25$

7. $10n^2 + 40n + 68$

$10(n^2 + 4n + 4) + 68 - 40$

$10(n + 2)^2 + 28$

8. $-3d^2 - 18d + 4$

$-3(d^2 + 6d + 9) + 4 + 27$

$-3(d + 3)^2 + 31$

9. $4y^2 + 48y - 15$

$4(y^2 + 12y + 36) - 15 - 144$

$4(y + 6)^2 - 159$

10. $9c^2 - 18c - 23$

$9(c^2 - 2c + 1) - 23 - 9$

$9(c - 1)^2 - 32$

Algebra 1	Name Answer Key___________
Unit 8	Date___________________
Lesson 7 Practice	Period__________________

1. For the expression, the first step to completing the square is shown.

$$x^2 + 18x + 56$$
$$(x^2 + 18x) + 56$$

1. Finish factoring the perfect square trinomial in vertex form.

$(x^2 + 18x + 81) + 56 - 81$

$(x + 9)^2 - 25$

In the expression, the value that was added to create the perfect square trinomial is being multiplied by ____1_____, so the value that should be subtracted from 56 to maintain equivalency is ___81_____.

Complete the table to complete the square without using algebra tiles.

	Process	**Equation**
	Original equation	$y = -5x^2 - 20x + 60$
2.	Factor out the common factor of the first two terms.	$y = \underline{-5}(x^2 + \underline{\ 4\ }x) + 60$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = \underline{-5}(x^2 + \underline{\ 4\ }x + \underline{\ 4\ }) + 60$
4.	Then, to maintain equivalence, multiply the factor by the value. Add the opposite of that value.	$y = \underline{-5}(x^2 + \underline{\ 4\ }x + \underline{\ 4\ }) + 60 + \underline{\ 20\ }$
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = \underline{-5}(x + \underline{\ 2\ })^2 + 60 + \underline{\ 20\ }$
6.	Combine like terms.	$y = \underline{-5}(x + \underline{\ 2\ })^2\ \underline{\ +\ }\ \underline{\ 80\ }$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 2x - 15$

$y = (x^2 - 2x + 1) - 15 - 1$

$y = (x - 1)^2 - 16$

8. Key Features

Key Features					
x-intercept		vertex	$(1, -16)$	domain	All Real #'s
x-intercept		increasing	$x > 1$	range	$y \geq -16$
y-intercept	$(0, -15)$	decreasing	$x < 1$	opens	Up
left end behavior		$x \to -\infty$, $y \to$ __∞__			
right end behavior		$x \to \infty$, $y \to$ __∞__			

9. $y = 2x^2 - 8x + 6$

$y = 2(x^2 - 4x + 4) + 6 - 8$

$y = 2(x - 2)^2 - 2$

10. Key Features

Key Features					
x-intercept		vertex	$(2, -2)$	domain	All Real #'s
x-intercept		increasing	$x > 2$	range	$y \geq -2$
y-intercept	$(0,6)$	decreasing	$x < 2$	opens	Up
left end behavior		$x \to -\infty$, $y \to$ __∞__			
right end behavior		$x \to \infty$, $y \to$ __∞__			

Algebra 1
Unit 8
Lesson 8 Practice

Name Answer Key____________
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The angle, in degrees, at which a batter hits a baseball can affect the distance the ball travels, in feet. The function $B(x) = -0.2x^2 + 14x - 21$ models the distance the baseball travels in ft. depending on the angle, in degrees, at which the ball is hit.

1. What does the variable x represent? Include the units.
   x represents the angles in degrees, at which the ball is hit.

2. What does the variable y represent? Include the units.
   y represents the distance the baseball travels in feet.

3. Interpret the y-intercept. Include units.
   The y-intercept is -21 feet. At an angle of $0°$, the baseball travels -21 feet.

4. Rewrite the function in vertex form. Show your work.
   $-0.2(x^2 - 70x + 1225) - 21 + 245$
   $-0.2(x - 35)^2 + 224$

5. Interpret the vertex in terms of the context. Include units.
   The vertex is $(35,224)$. At an angle of $35°$, the baseball travels 224 feet.

6. Complete the table of values for $M(x)$.

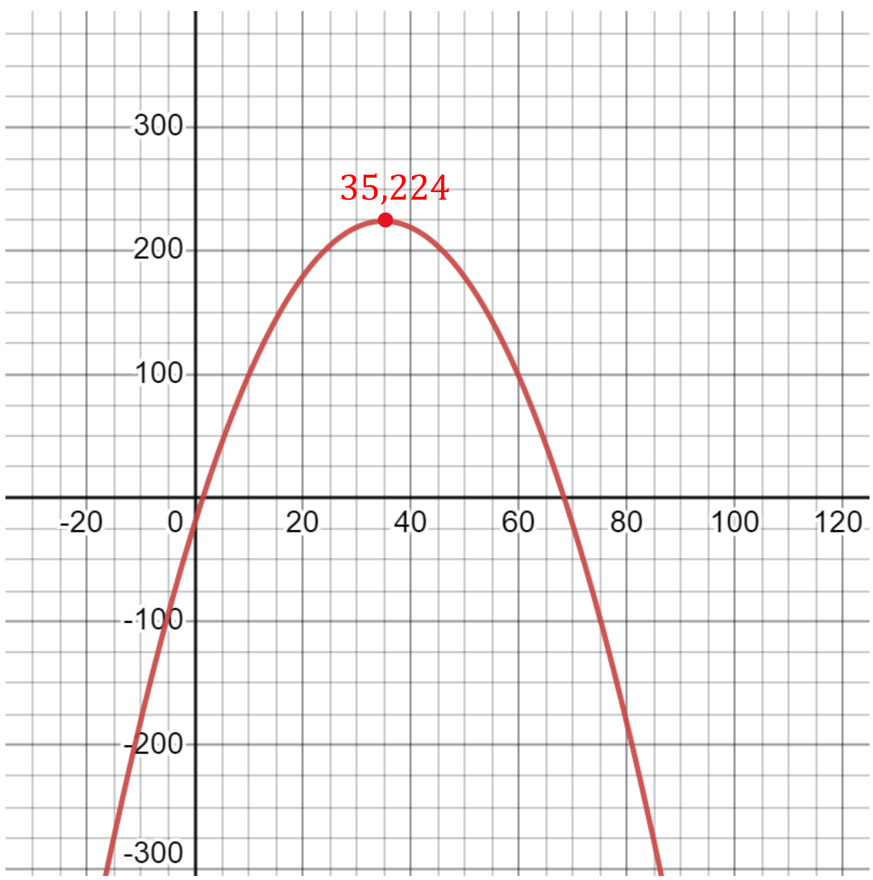
x	-2	0	2	7	17	23	35	68
$M(x)$	-49.8	-21	6.2	67.2	159.2	195.2	224	6.2

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x-intercept		vertex	$(35,224)$	domain	$\mathbb{R}$
x-intercept		increasing	$x < 35$	range	$y \leq 224$
y-intercept	$(0,-21)$	decreasing	$x > 35$	opens	Down
left end behavior	$x \to -\infty$, $y \to$ $-\infty$				
right end behavior	$x \to \infty$, $y \to$ $-\infty$				

9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.
$\{x | 1 \leq x \leq 69\}$ using estimated values from table.

Algebra 1
Unit 8
Lesson 9 Practice

Name_Answer Key________
Date___________________
Period__________________

Florida Real Estate Team just bought a 100 unit apartment building and plans to rent all of the apartments out. The function $P(u) = -10u^2 + 1760u - 50{,}000$ represents the profit earned, P, depending on the amount of apartment units, u, they rent.

1. Rewrite the function in vertex form.
   $P(u) = -10(u^2 - 176 + 7{,}744)^2 - 50{,}000 + 77{,}400$
   $P(u) = -10(u - 88)^2 + 27{,}440$

Complete the statements.

2. The
   - o vertex
   - ● factored
   - o standard

   form of the function could be used to determine the interval on which the real estate company would experience a profit.

3. The
   - ● vertex
   - o factored
   - o standard

   form of the function can be used to determine when the real estate company experiences the greatest profit.

4. The company is predicted to experience the greatest profit when ___88___ units are rented, resulting in a profit of $___27,440___.

Adam is coding a new game where darts are launched at a moving target. He enters the function $H(t) = -0.5t^2 + 6t - 10$ to model the height, H, of the darts in relationship to the ground after t seconds.

5. Rewrite the function in vertex form.

$H(t) = -0.5(t^2 - 12 + 36)^2 - 10 + 18$

$H(t) = -0.5(t - 6)^2 + 8$

6. Rewrite the function in factored form.

$H(t) = -0.5(t^2 + 12t + 20)$

$H(t) = -0.5(t - 2)(t - 10)$

Complete the statements.

7. The

- o vertex
- ● factored
- o standard

form of the function can be used to determine how long the dart is above the ground. The dart is in the air for $2 < t < 10$, 8 seconds___.

8. The dart was launched from a point __10___ units

- o above
- ● below

the ground and went

- ● above
- o below

ground after __2___ seconds.

9. The

- ● vertex
- o factored
- o standard

form of the function can be used to determine when the dart reaches its highest point.

10. The dart reaches its highest point of __8___ units __6___ seconds after it is launched.